Academic Stress Analysis and Prediction among IIT Students using Machine Learning

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Abstract

Academic stress is an important issue among college students, especially in competitive academic environments such as IITs. This project studies academic stress using primary survey data collected through Google Forms. The survey includes academic, lifestyle, and anxiety-related factors such as sleep hours, screen time, exercise frequency, diet quality, CGPA, study hours, and GAD-based anxiety indicators. Exploratory Data Analysis was performed to identify important patterns, followed by machine learning models for predicting student stress levels. Logistic Regression achieved the best accuracy of 71.4%, outperforming Random Forest and Decision Tree models. The results show that lifestyle factors such as sleep, diet, and exercise have a strong influence on stress, while anxiety indicators are closely related to high stress levels.

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1 Introduction

Academic stress is a common problem among students due to continuous assignments, examinations, academic competition, career pressure, and personal expectations. In highly competitive institutions, students often face pressure to maintain good academic performance while also managing lifestyle habits and mental health.

The main objective of this project is to analyze academic stress among IIT students and identify the major factors affecting it. The project also aims to build a machine learning model that can predict whether a student belongs to a Low, Medium, or High stress category.

The work is divided into four major parts:

- Data collection through Google Forms
- Data cleaning and preprocessing
- Exploratory Data Analysis
- Machine learning model building and deployment using Streamlit

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2 Dataset Description

The dataset was collected through a Google Form survey. A total of 102 responses were collected from students. Since the data was collected directly from students, it is considered primary data.

The dataset contains different categories of features:

2.1 Demographic Features

These features describe the background of the student.

- Gender
- Branch
- Year of study

2.2 Academic Features

These features represent academic performance and workload.

- CGPA range
- Daily study hours

2.3 Lifestyle Features

These features represent daily habits of students.

- Sleep hours
- Screen time
- Exercise frequency
- Diet quality

2.4 Mental Health Indicators

Two GAD-related indicators were used:

- Feeling nervous or anxious
- Feeling irritable or restless

A combined GAD score was calculated using:

$$GAD_Score = \frac{GAD_Nervous + GAD_Irritable}{2}$$

This score helps represent anxiety-related symptoms numerically.

3 Target Variable Creation

The original stress score was collected on a scale from 1 to 10. For classification, this continuous stress score was converted into three categories:

- Low Stress: score 1–3
- Medium Stress: score 4–6
- High Stress: score 7–10

This transformation makes the problem suitable for multi-class classification.

$$Stress_Level = \begin{cases} Low, & Stress \leq 3 \\ Medium, & 4 \leq Stress \leq 6 \\ High, & Stress \geq 7 \end{cases}$$

4 Data Cleaning and Preprocessing

Before applying machine learning models, the dataset was cleaned and transformed.

4.1 Handling Missing Values

Machine learning models such as Logistic Regression cannot directly handle missing values. Therefore, missing values were checked and handled using suitable methods such as dropping incomplete records or filling missing values.

4.2 Encoding Categorical Variables

Most features such as sleep hours, screen time, exercise frequency, diet quality, and CGPA were categorical. These were converted into numerical form so that machine learning models could process them.

For example:

- Less than 4 hours sleep = 1
- 4–6 hours sleep = 2
- 6–8 hours sleep = 3
- More than 8 hours sleep = 4

4.3 Feature Engineering

New useful features were created from existing columns. The most important engineered feature was the GAD score, which combines two anxiety-related indicators.

Feature engineering is important because good features can improve model performance more than simply using complex models.

5 Exploratory Data Analysis

Exploratory Data Analysis was performed to understand patterns in the data and identify important stress-related factors.

5.1 Overall Stress Distribution

The stress distribution showed that a significant number of students fall under the High stress category. Around 41% of respondents reported high stress levels. This indicates that academic stress is a serious issue among the surveyed students.

5.2 Sleep Hours vs Stress

The analysis shows a clear inverse relationship between sleep and stress. Students sleeping less than 4 hours generally have very high stress levels. Students sleeping 6–8 hours or more than 8 hours show lower stress levels.

This suggests that sleep is one of the strongest lifestyle factors affecting stress. Lack of sleep may reduce concentration, emotional stability, and academic performance, which can increase stress.

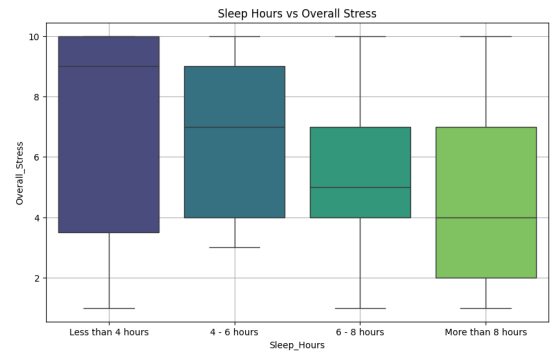

Figure 1: Sleep Hours vs Overall Stress

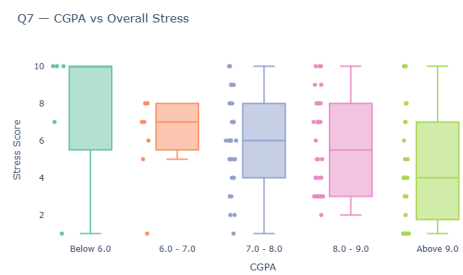

Figure 2: CGPA vs Overall Stress

5.3 CGPA vs Stress

The CGPA analysis shows that students with lower CGPA generally experience higher stress. Students with higher CGPA, especially above 9.0, tend to show lower median stress.

However, the relationship is not perfect. Some high-CGPA students also report high stress, possibly due to pressure to maintain academic performance, competition, and expectations.

5.4 Branch vs Stress

Branch-wise analysis shows that stress levels are broadly similar across branches. Some branches show slightly higher variation, but no branch is completely stress-free.

This suggests that stress is not only dependent on academic branch. Lifestyle, personal habits, and psychological factors may play a bigger role.

5.5 GAD Score vs Stress

The GAD score increases as stress level increases. Low stress students mostly have lower GAD scores, while high stress students show higher anxiety-related scores.

This indicates a strong positive relationship between stress and anxiety symptoms.

$$Stress \uparrow \Rightarrow GAD_Score \uparrow$$

This is one of the strongest findings of the analysis.

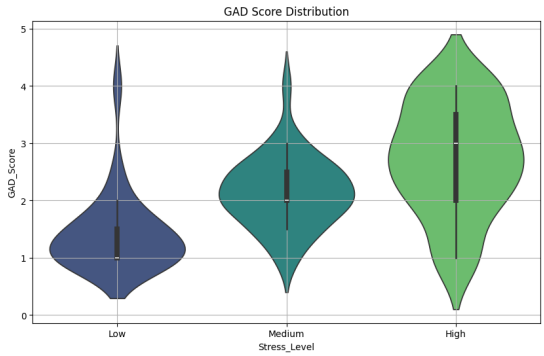

Figure 3: GAD Score Distribution across Stress Levels

5.6 Lifestyle Factors vs Stress

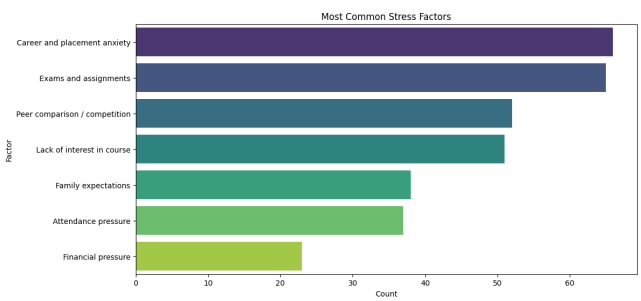

Figure 4: Lifestyle Factors vs Overall Stress

Several lifestyle factors were analyzed together.

5.6.1 Screen Time. Students with higher non-academic screen time generally show higher stress. Excessive screen time may affect sleep, productivity, and mental health.

5.6.2 Exercise Frequency. Students who exercise daily or regularly tend to have lower stress. Exercise can act as a stress-reducing factor by improving mood, sleep, and physical health.

5.6.3 Diet Quality. Students with better diet quality generally report lower stress. Poor diet and irregular eating habits may negatively affect energy levels and emotional balance.

5.6.4 Study Hours. Study hours do not show a simple linear relationship. Very low study hours may cause stress due to lack of preparation, while very high study hours may also increase stress due to workload and burnout.

Thus, balanced study habits appear more useful than extreme study patterns.

6 Machine Learning Methodology

The goal of machine learning was to predict the stress level of a student as Low, Medium, or High based on lifestyle, academic, and anxiety-related features.

The models used were:

- Logistic Regression
- Random Forest
- Decision Tree

The data was divided into training and testing sets. The training set was used to train the model, while the testing set was used to evaluate performance on unseen data.

7 Logistic Regression

Logistic Regression is a classification algorithm used to predict categorical outcomes. For multi-class classification, it estimates the probability of each class and selects the class with the highest probability.

The softmax form of Logistic Regression is:

$$P(y = k|x) = \frac{e^{\beta_k^T x}}{\sum_j e^{\beta_j^T x}}$$

where:

- x is the input feature vector
- β_k represents model parameters for class k
- $P(y = k|x)$ is the probability of class k

Logistic Regression assumes that classes can be separated using linear decision boundaries. In this project, Logistic Regression performed best because the dataset is small and many relationships are reasonably simple.

8 Random Forest

Random Forest is an ensemble learning method that combines multiple decision trees. Each tree makes a prediction, and the final prediction is decided by majority voting.

$$\hat{y} = majority\ vote(T_1, T_2, ..., T_n)$$

Random Forest can capture non-linear relationships better than Logistic Regression. However, on small datasets, it may overfit noise and not generalize well.

In this project, Random Forest performed well for High stress detection but had lower overall accuracy than Logistic Regression.

9 Decision Tree

Decision Tree is a simple model that splits the data based on feature conditions. It is easy to interpret but can overfit easily, especially when the dataset is small.

Decision Tree gave the lowest accuracy among the tested models.

10 Model Evaluation Metrics

Several evaluation metrics were used.

10.1 Accuracy

Accuracy measures the percentage of correct predictions.

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

10.2 Precision

Precision tells how many predicted positives were actually correct.

$$Precision = \frac{TP}{TP + FP}$$

10.3 Recall

Recall tells how many actual positives were correctly identified.

$$Recall = \frac{TP}{TP + FN}$$

10.4 F1 Score

F1 Score is the harmonic mean of precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

F1 score is useful when class distribution is not perfectly balanced.

11 Results

11.1 Model Comparison

Model	Accuracy	F1 Score
Logistic Regression	71.4%	0.70
Random Forest	66.7%	0.65
Decision Tree	61.9%	0.64

Table 1: Model Performance Comparison

Logistic Regression achieved the highest accuracy of 71.4%. This means that it correctly predicted around 71 out of every 100 stress categories.

11.2 Logistic Regression Result

The confusion matrix for Logistic Regression was:

$$\begin{bmatrix} 7 & 1 & 1 \\ 0 & 5 & 0 \\ 2 & 2 & 3 \end{bmatrix}$$

The model performs well for High and Low stress classes. However, it struggles with Medium stress. This is because Medium stress lies between Low and High stress, so its feature patterns overlap with both classes.

11.3 Random Forest Result

The confusion matrix for Random Forest was:

$$\begin{bmatrix} 8 & 1 & 0 \\ 0 & 3 & 2 \\ 3 & 1 & 3 \end{bmatrix}$$

Random Forest correctly identified most High stress cases, but it performed worse overall compared to Logistic Regression. This may be because the dataset size is small, and Random Forest can overfit small datasets.

11.4 Important Observation

Both Logistic Regression and Random Forest struggle with Medium stress classification. This suggests that stress is not naturally divided into three clear categories. Instead, stress behaves more like a continuous psychological state.

12 Discussion

The results show that lifestyle habits have a strong influence on stress. Sleep, diet, exercise, and anxiety indicators are more useful predictors than branch or CGPA alone.

The strongest pattern observed is:

$$Lifestyle \rightarrow Stress \rightarrow Anxiety$$

Students with poor lifestyle habits tend to report higher stress, and students with higher stress also show higher GAD scores.

Another important finding is that academic branch does not strongly separate stress levels. This means stress is more individual and lifestyle-based rather than purely branch-based.

13 Streamlit Application

A Streamlit-based web application was developed to make the project interactive and practically usable. The application is structured into multiple sections, including Overview, Exploratory Data Analysis (EDA), Model Results, and a Stress Predictor interface.

The application allows users to:

- Visualize key insights from the survey data
- Explore relationships between lifestyle, academic, and mental health factors
- Compare the performance of different machine learning models
- Predict stress level based on user-provided inputs

The application loads the cleaned dataset, processes user inputs, computes derived features such as the GAD score, and uses a trained machine learning model to predict the stress level. It also provides interactive visualizations and an intuitive user interface for better understanding of the results.

Application Link: ykfes2czkq9apprqwsnyq8e.streamlit.app

14 Limitations

This project has some limitations:

- Dataset size is small with only 102 responses
- Data is self-reported, so responses may contain bias
- Most respondents are from one academic year, so the dataset is not fully balanced
- Stress is complex and cannot be completely captured by limited survey questions
- Medium stress category overlaps with Low and High categories

15 Future Work

The project can be improved in the future by:

- Collecting more responses from different years and branches
- Adding more psychological and academic features
- Using advanced models such as XGBoost or LightGBM
- Applying hyperparameter tuning

- Using regression to predict stress score directly instead of classifying it into categories
- Adding personalized recommendations in the app

16 Conclusion

This project analyzed academic stress among IIT students using survey data, exploratory data analysis, machine learning, and an interactive Streamlit application.

The major findings are:

- Students with less sleep generally have higher stress
- Better diet and regular exercise are linked with lower stress
- Higher GAD scores are strongly associated with higher stress
- CGPA shows some relationship with stress, but it is not the only factor
- Logistic Regression performed best with 71.4% accuracy
- Medium stress is difficult to classify because stress exists on a continuum

Overall, the study shows that lifestyle and mental health indicators are important predictors of student stress. The project also demonstrates how machine learning can be used to build a practical stress prediction system.

17 References

- Scikit-learn Documentation
- Streamlit Documentation
- GAD-7 Anxiety Scale Literature
- Student Mental Health and Academic Stress Studies